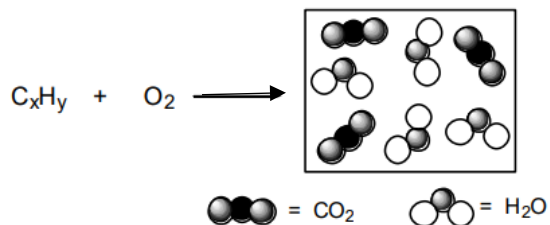


Unit 1 MC Practice

1. 2.500 grams of $\text{MgSO}_4 \cdot x \text{H}_2\text{O}$, a hydrated salt with an unknown water content, is dried in an oven to constant mass until all water has been removed. After drying, the anhydrous salt has a mass of 1.221 grams. How many moles of water are present per mole of hydrated magnesium sulfate? (FW of $\text{MgSO}_4 = 120 \text{ g/mol}$)
- a. 1 b. 3 c. 5 d. 7
3. When hafnium metal is heated in an atmosphere of chlorine gas, the product of the reaction is found to contain 62.2 percent Hf by mass and 37.4 percent Cl by mass. What is the empirical formula for this compound?
- a. Hf_2Cl_3 b. HfCl_2 c. HfCl_3 d. HfCl_4
4. Which of the following represents the correct method for converting 11.0 g of copper metal to the equivalent number of copper atoms?
- a. $11 \left(\frac{1}{63.55} \right) \left(\frac{6.02 \times 10^{23}}{1} \right)$
- b. $11 \left(\frac{1}{63.55} \right)$
- c. $11 \left(\frac{1}{63.55} \right) \left(\frac{63.55}{6.02 \times 10^{23}} \right)$
- d. $11 \left(\frac{63.55}{1} \right) \left(\frac{6.02 \times 10^{23}}{1} \right)$

5. A hydrocarbon of unknown formula C_xH_y was submitted to combustion analysis: the results are shown in the diagram below. What is the empirical formula of the compound?



- a. CH b. CH_2 c. C_3H_4 d. C_3H_8
6. How many grams of calcium nitrate, $Ca(NO_3)_2$, contains 48 grams of oxygen atoms?
- a. 82 g b. 120 g c. 190 g d. 320 g

8. Which IUPAC name/formula combination is **wrong**?

- a. chlorous acid/ $HClO_2$ c. ammonium permanganate/ NH_4MnO_4
 b. dicarbon tetranitride/ C_2N_4 d. copper (II) periodate/ $CuIO_4$

9. Which set of molecules is in order from lowest to highest percent mass of oxygen?

- a. $CH_3OH < CH_3CH_2OH < CH_3CH_2CH_2OH < HOCH_2CH_2OH$
 b. $HOCH_2CH_2OH < CH_3OH < CH_3CH_2OH < CH_3CH_2CH_2OH$
 c. $CH_3CH_2CH_2OH < CH_3CH_2OH < CH_3OH < HOCH_2CH_2OH$
 d. $CH_3CH_2CH_2OH < CH_3CH_2OH < HOCH_2CH_2OH < CH_3OH$

11. What pressure (in atm) would be exerted by 76 g of fluorine gas in a 1.50 liter vessel at -37°C ?

- a. 4.1 atm b. 8.2 atm c. 26 atm d. 84 atm

12. Which of the following statements is true?

- I. The molar mass of CaCO_3 is 100.1 g mol^{-1} .
II. 50 g of CaCO_3 contains about 9×10^{23} oxygen atoms.
III. A 200 g sample of CaCO_3 contains about 2 moles of CaCO_3 .

- a. I only b. II only c. I and III only d. I, II, and III

13. In which of the following compounds is the mass ratio of element X to oxygen closest to 2.5 to 1? (The molar mass of X is 40.0 g/mol .)

- a. X_3O_2 b. X_2O c. XO_2 d. XO

14. If 87 grams of K_2SO_4 (molar mass 174 g/mol) is dissolved in enough water to make 250 mL of solution, what are the concentrations of the potassium and the sulfate ions?

$[\text{K}^+]$	$[\text{SO}_4^{2-}]$
----------------	----------------------

- | | |
|----------------------|-------------------|
| a. 0.020 M | 0.020 M |
| b. 1.0 M | 2.0 M |
| c. 2.0 M | 1.0 M |
| d. 4.0 M | 2.0 M |

21. An oxide of lead contains 90.65% pb by weight. The empirical formula is:

- a. Pb b. PbO c. Pb₃O₄ d. PbO₂

22. The mass in grams of 2.6×10^{22} chlorine atoms is:

- a. 0.76 g b. 1.5 g c. 3.2 g d. 4.4 g

23. Identify the INCORRECT statement.

- a. Helium in a balloon: an element c. Tap water: a compound
b. Paint: a mixture d. Mercury in a barometer: an element

24. Which one of the following samples contains the most **atoms**?

- a. 1 mol of UF₆(g) c. 1 mol of CH₃COCH₃(l)
b. 1 mol of He(g) d. all contain the same number of atoms

26. The neutral atoms of all of the isotopes of the same element have:

- a. different numbers of protons c. the same number of electrons
b. equal numbers of neutrons d. the same mass number

27. Which one of the following samples contains the most **molecules**?

- a. 1 mol of $\text{UF}_6(\text{g})$
- c. 1 mol of $\text{CH}_3\text{COCH}_3(\text{l})$
- b. 1 mol of $\text{He}(\text{g})$
- d. all contain the same number of atoms

28. Analysis of a sample of a covalent compound showed that it contained 14.4% hydrogen and 85.6% carbon by mass. What is the empirical formula of the compound?

- a. CH
- b. CH_2
- c. CH_3
- d. C_2H_3

29. What mass of cerussite, PbCO_3 , would contain 35.0 grams of lead?

- a. 27.1 g
- b. 35.6 g
- c. 45.1 g
- d. 51.7 g

31. The simplest formula for an oxide of nitrogen that is 36.8 percent nitrogen by weight is:

- a. N_2O
- b. NO
- c. N_2O_3
- d. NO_2

32. How many aluminum atoms are there in 3.50 g of Al_2O_3 ?

- a. 2.07×10^{22}
- b. 2.07×10^{23}
- c. 4.13×10^{22}
- d. 4.13×10^{23}

33. The density of chlorine gas at STP, in grams per liter, is approximately:

- a. 1.3 b. 3.2 c. 4.5 d. 6.2

37. Consider the species, ^{72}Zn , ^{75}As , and ^{74}Ge . These species have:

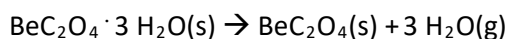
- a. the same number of electrons c. the same number of protons
b. the same number of neutrons d. the same mass number

AP Free Response Practice #1 [10 points, 1991 Form B]

1. Answer the following questions about BeC_2O_4 and its hydrate.

a. Calculate the mass percent of carbon in the hydrated solid with the formula $\text{BeC}_2\text{O}_4 \cdot 3 \text{H}_2\text{O}$. (2 points)

b. When heated to $220.^\circ\text{C}$, $\text{BeC}_2\text{O}_4 \cdot 3 \text{H}_2\text{O}$ dehydrates completely as represented below:



If 3.21 g of $\text{BeC}_2\text{O}_4 \cdot 3 \text{H}_2\text{O}$ is heated to $220.^\circ\text{C}$, calculate each of the following:

i. The mass of BeC_2O_4 formed. (1 point)

ii. The volume of $\text{H}_2\text{O}(\text{g})$ released, measured at $220.^\circ\text{C}$ and 735 mmHg. (2 points)

A student repeats the dehydration from part (b) in an attempt to experimentally determine the number of moles of water in one mole of $\text{BeC}_2\text{O}_4 \cdot 3 \text{H}_2\text{O}$. The student collects the data shown in the table below.

Mass of empty crucible	36.48 g
Initial mass of sample and crucible	39.69 g
Mass of sample and crucible after first heating	38.82 g

c. Use the data above to:

i. Calculate the total number of moles of water lost when the sample was heated. (1 point)

- d. Is the student's experimentally determined waters of hydration greater than, less than, or equal to the waters of hydration in the accepted formula? Provide a reasonable explanation for error and how this error affected the student's results. (2 points)

AP Free Response Practice #2 [6 points]

2. A 7.96 mg sample of a compound containing the elements, C, H and S is burned in oxygen and found to form 16.65 mg of CO_2 . The sulfur in 4.31 mg of the compound is converted into sulfate by a series of reactions and precipitated as BaSO_4 . The BaSO_4 was found to have a mass of 11.96 mg.

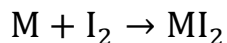
a. What is the mass of carbon found in the original sample? [1 point]

b. How many moles of sulfur were present in the original sample? [1 point]

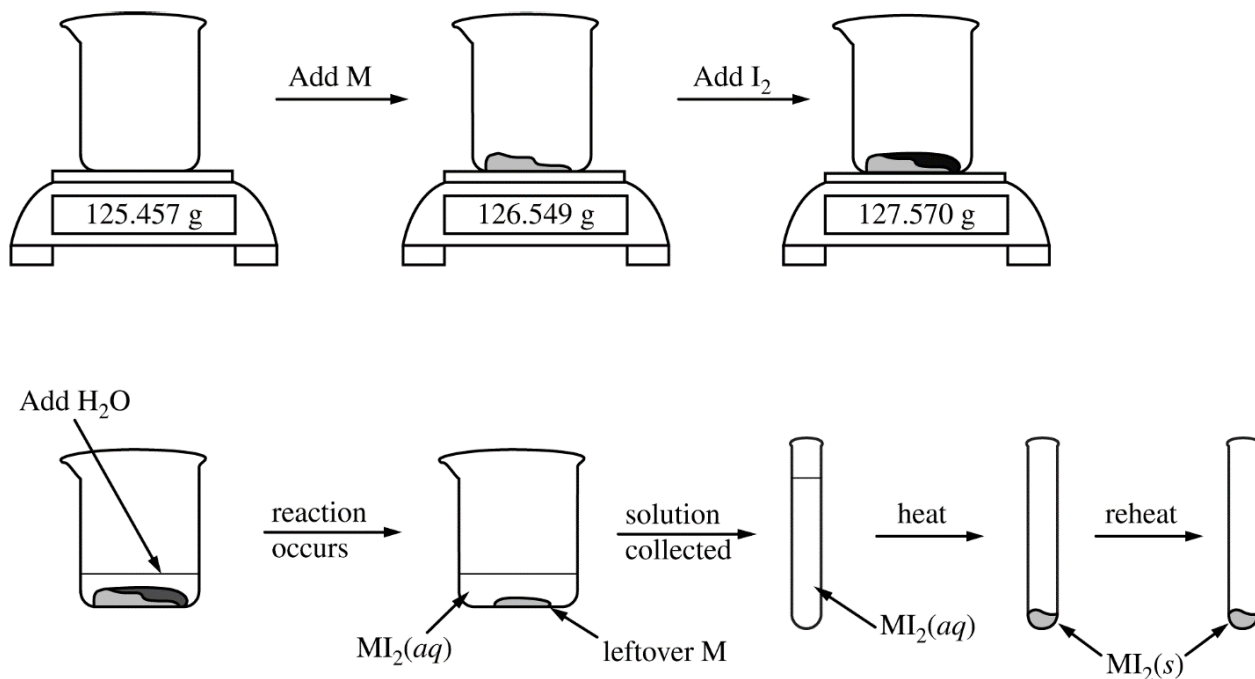
c. Using your calculations, determine the empirical formula of the compound. [2 points]

d. When the BaSO_4 was precipitated and filtered from solution, some of the BaSO_4 passed through the filter paper and wasn't collected. How will this lab error affect the calculated empirical formula? [2 points]

AP Free Response Practice #3 (2016, modified) [10 points]



3. To determine the molar mass of an unknown metal, M, a student reacts iodine with an excess of the metal to form the water-soluble compound MI_2 , as represented by the equation above. The reaction proceeds until all of the I_2 is consumed. The $\text{MI}_2(\text{aq})$ solution is quantitatively collected and heated to remove the water, and the product is dried and weighed to constant mass. The experimental steps are represented below, followed by a data table.



Data for Unknown Metal Lab	
Mass of beaker	125.457 g
Mass of beaker + metal M	126.549 g
Mass of beaker + metal M + I_2	127.570 g
Mass of MI_2 , first weighing	1.284 g
Mass of MI_2 , second weighing	1.284 g

- a) Given that the metal M is in excess, calculate the number of moles of I_2 that reacted. [2 points]

- b) Determine the following for the unknown metal M.
- Calculate the molar mass of the unknown metal M. [2 points]
 - What is the most likely identity of the unknown metal M? [1 point]
 - Calculate the number of moles of unknown metal M that reacted. [2 points]
- c) Provide a calculation to confirm the empirical formula of the compound MI_2 based on the data shown. [2 points]
- d) If the student failed to heat to constant mass, would the calculated molar mass of the unknown metal M be greater than, less than, or equal to the actual molar mass? Explain. [1 point]

AP Free Response Practice #4 [10 points]

4. Lysine is an amino acid which has the following elemental composition: C, H, O, N. It is found in the protein of foods such as beans, cheese, yogurt, meat, milk, brewer's yeast, wheat germ, and other animal proteins. The average 70 kg human needs 800 – 3,000 mg of lysine daily. In one experiment, 2.175 g of lysine was combusted to produce 3.94 g of CO_2 and 1.89 g H_2O . The molar mass of lysine is approximately 150 g/mol.

a) Determine the mass, in grams, of each of the following in the 2.175 g sample of lysine.

i. carbon [1 point]

ii. hydrogen [1 point]

b) In a separate experiment, 1.873 g of lysine was burned to produce 0.436 g of NH_3 .

i. Determine the mass, in grams, of N in the 1.873 g sample of lysine. [1 point]

ii. Determine mass percent of each element in lysine. [2 points]

c) Determine the mass, in grams, of O in the original 2.175 g sample of lysine. [1 *point*]

d) Using information derived from the provided data, determine the empirical formula of lysine. [3 *points*]

e) Determine the molecular formula of lysine. [1 *point*]